

Internet of Things:

Pay more attention to the secure storage of data collected

If the Internet of Things (IoT) is campaigning to run nearly every aspect of our digital lives, considerations need to be made to ensure a seamless and safe introduction, says Kevin Curran, IEEE senior member and reader in computer science at Ulster University, Ireland, as he talks to Dilin Anand of EFY about the challenges in implementing the IoT on a large scale



KEVIN CURRAN
IEEE SENIOR MEMBER AND
READER IN COMPUTER SCIENCE,
ULSTER UNIVERSITY, IRELAND

Q. What are the challenges in implementing the IoT?

A. Technical challenges include government regulations with regards to spectrum allocation, security, battery issues, cost and privacy. Security, standards and overburdening the network are three requirements that need to be focused on before implementing mass adoption.

Q. What are the little things that we easily overlook when dealing with the IoT scenario?

A. Good throughput is important for the IoT, but there will be trade-off between data speed and battery life. Multiple input multiple output (MiMo) technology is set to be a key part of these efficiency measures. Existing IoT sensors are not equipped to take advantage of 5G. New devices may need to have multiple antennae for fewer dropouts.

Q. What about standardisation?

A. We are still a long way from creating scalable solutions where products from different vendors can communicate with each other. Industrial industries might be the first to achieve some form of integrated IoT solutions because business value (like costs and preventative maintenance) can be measured and quantified.

Q. Are there any leading standards for the IoT domain?

A. One attempt to standardise is AllSeen Alliance (www.allseenalliance.org), which is an IoT industry group to encourage interoperability among connected devices, regardless of their manufacturer. The group's software will be based on AllJoyn, which is Qualcomm's open source IoT software.

Newcomers to this alliance include Automatic and Nest. Microsoft Research's Lab of Things software also attempts to simplify the monitoring, automating and controlling of all kinds of smart devices in homes. Then there is OpenRemote's open source software that can connect and automate all kinds of devices.

Q. What skills are needed for handling these devices?

A. It is difficult, and will remain so, to hire the skills required to manage IoT deployments as the actual devices, communications infrastructure and backend platforms grow in complexity. When the IoT takes over most parts

of the society, new challenges will arise. For instance, engineers will need to estimate proper wireless coverage. Over-provide and they waste an organisation's money.

Related is the challenging area of testing performance. Application performance needs testing. There will also be a need to be able to work out if sensor data spikes are being dealt with effectively. In general, network performance monitoring will often be the primary source of highlighting problems.

Q. How does one handle big data?

A. An IoT scenario deals with big volumes of data due to the large number of sensors involved. There are three main problems that must be solved: resolution, sensitivity and reliability. Compressed sensing involves reducing the number of samples collected in an IoT wireless sensor network. Thus, it is possible to create standalone applications that require fewer resources.

Q. What is your advice to IoT device manufacturers?

A. They need to consider the correct forms of cryptographic algorithms and modes needed for IoT devices. An international organisation for standardisation/international electrotechnical commission (ISO/IEC) 29192 standard was devised to implement lightweight cryptography on constrained devices.

Q. What is the one piece of advice for all IoT companies?

A. They will have to pay more attention to the secure storage of data collected via the IoT as legal repercussions creep in and data being collected increases. This data is generally stored in the cloud. So, they should, in particular with large data sets, pay extra attention to the data lifecycle phases and ensure that aspects such as data destruction are provided and auditable as part of the service.

Q. What is the most critical element to ensure security for data intensive applications like the IoT?

A. It is critical that companies implement a layered security strategy regarding cloud services, as their data is more exposed now. It is critical to get buy-in from upper management. Also, security breaches can greatly affect their reputation. ●